English

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ALLEN® CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023-2024)

Test Pattern

NEET (UG)

MINOR

14-08-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE - MEA & MEPS-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to **fill 0012345678.**

Test Booklet Code

E7

This Booklet contains **28** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II). 50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
   (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
   (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
   Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720**.
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : ____________________

Form Number : in figures ____________________

: in words ____________________

Centre of Examination (in Capitals) : ____________________

Candidate's Signature : ____________________ **Invigilator's Signature :** ____________________

Your Target is to secure Good Rank in Pre-Medical 2024

SECTION-A (PHYSICS)

1. The de-Broglie wavelength of a neutron at 27°C is λ. What will be its wavelength at 927°C ?

(1) $\lambda/2$　(2) $\lambda/3$　(3) $\lambda/4$　(4) $\lambda/9$

2. A nucleus ruptures into two nuclear parts which have their velocity ratio equal to 2 : 1. What will be the ratio of their nuclear size (nuclear radius)

(1) $2^{1/3} : 1$　(2) $1 : 2^{1/3}$

(3) $3^{1/2} : 1$　(4) $1 : 3^{1/2}$

3. Let F_{pp}, F_{PN} and F_{NN} denote the nuclear force between proton-proton, proton-neutron and neutron-neutron pair respectively.When separation is 1 fm :-

(1) $F_{pp} < F_{PN} = F_{NN}$

(2) $F_{pp} > F_{PN} = F_{NN}$

(3) $F_{pp} = F_{PN} = F_{NN}$

(4) $F_{pp} < F_{PN} < F_{NN}$

4. The binding energy per nucleon in deuterium and helium nuclei are 1.1 MeV and 7.0 MeV, respectively. When two deuterium nuclei fuse to form a helium nucleus the energy released in the fusion is :-

(1) 2.2 MeV　(2) 28.0 MeV

(3) 30.2 MeV　(4) 23.6 MeV

5. A nucleus with atomic number Z and neutron number N undergoes two decay processes. The result is a nucleus with atomic number Z – 3 and neutron number N – 1, which decay processes took place ?

(1) Two β^--decays

(2) Two β^+-decay

(3) An α-decay and a β^--decay

(4) An α-decay and a β^+-decay

6. Half life of a radio-active substance is 20 minutes. The time between 20% and 80% decay will be :-

(1) 20 minutes　(2) 40 minutes

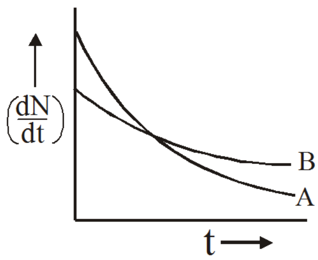
(3) 30 minutes　(4) 25 minutes

7. A radioactive sample at any instant has its disintegration rate 5000 disintegrations per minute. After 5 minutes, the rate is 1250 disintegrations per minute. Then, the decay constant (per minute) is :-

(1) $0.8\ \ell n\ 2$　(2) $0.4\ \ell n\ 2$

(3) $0.2\ \ell n\ 2$　(4) $0.1\ \ell n\ 2$

8. The variation of decay rate of two radioactive samples A and B with time is shown in figure. Which of the following statements is true ?



(1) Decay constant of A is greater than that of B, hence A always decays faster than B

(2) Decay constant of B is greater than that of A but its decay rate is always smaller than that of A

(3) Decay constant of A is greater than that of B but it does not always decay faster than B

(4) All of the above

9. The half-life of a radioactive substance against α-decay is 1.2×10^7 s. What is the decay rate for 4×10^{15} atoms of the substance :-

(1) 4.6×10^{12} atom / s

(2) 2.3×10^{11} atom / s

(3) 4.6×10^{10} atom / s

(4) 2.3×10^{8} atom / s

10. If the X-ray tube is working at 25 kV then the minimum wavelength of X-rays will be :-

(1) 0.49Å (2) 0.29Å

(3) 0.19Å (4) 0.39Å

11. **Assertion :** X-rays cannot be obtained in the emission spectrum of hydrogen atom.

Reason : Maximum energy of photons emitted from hydrogen spectrum is 13.6 eV.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

12. An X-ray tube is operated at 18 kV. The maximum velocity of electron striking the target is

(1) 8×10^7 m/s (2) 6×10^7 m/s

(3) 5×10^7 m/s (4) None of these

13. When a string fixed at its both end vibrate in 1 loop, 2 loops, 3 loops and 4 loops. The frequency are in ratio of :

(1) 1 : 1 : 1 : 1 (2) 1 : 2 : 3 : 4

(3) 4 : 3 : 2 : 1 (4) 1 : 4 : 9 : 16

14. The frequency of a turning fork A is 2% more than the frequency of a standard turning fork. The frequency of another fork B is 3% less than the frequency of a standard turning fork. If 6 beat/s are heard when the two turning forks A and B are excited, the frequency A is

(1) 120 Hz (2) 122.4 Hz

(3) 116.4 Hz (4) 130 Hz

15. **Assertion :** In the experiment of finding speed of sound by resonance tube method, as the level of water is lowered, wavelength increases.

Reason : By lowering the water level number of loops increases.

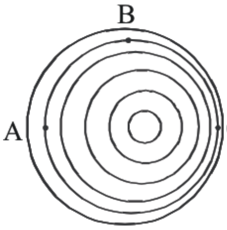
(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

16. Three observers, A, B, and C are listening to a moving source of sound. The diagram below shows the location of the wavecrests of the moving source with respect to the three stationary observers. Which of the following is true?



(1) The wavefronts move faster at C than at A and B.

(2) The frequency of the sound is highest at A.

(3) The frequency of the sound is highest at B.

(4) The frequency of the sound is highest at C.

17. An open pipe is suddenly closed with the result that the second overtone of the closed pipe is found to be higher in frequency by 100Hz than the first overtone of the original pipe the fundamental frequency of open pipe will be :-

(1) 100 Hz (2) 300 Hz

(3) 150 Hz (4) 200 Hz

18. Tuning fork A produces 4 beats/sec. with another tuning fork B of frequency 288 Hz. If fork is loaded with little wax, no. of beats per sec decreases. The frequency of the fork A, before loading is

(1) 290 Hz (2) 288 Hz

(3) 292 Hz (4) 284 Hz

19. Calculate the speed of sound in a liquid of density 2000 kg/m^3 and bulk modulus 2×10^9 N/m^2.

(1) 500 m/s

(2) 1000 m/s

(3) 125 m/s

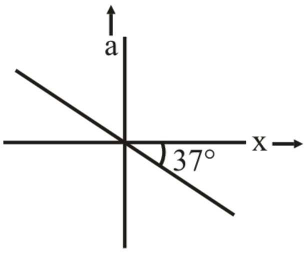
(4) 250 m/s

20. The mass of a 4 meter wire is 0.01 kg, and it is stretched by a force of 400N. What is the speed of transverse wave in the wire :-

(1) 400 m/sec (2) 200 m/sec

(3) 800 m/sec (4) None

21. From the given a-x graph. Find time period of SHM :-



(1) $\frac{4\pi}{\sqrt{3}}$ (2) $\frac{2\pi}{\sqrt{3}}$

(3) $\frac{\sqrt{3}\pi}{4}$ (4) $\frac{\sqrt{3}\pi}{2}$

22. PE of a particle in SHM is 7.5 J at $x = \frac{A}{2}$. Find its total energy.

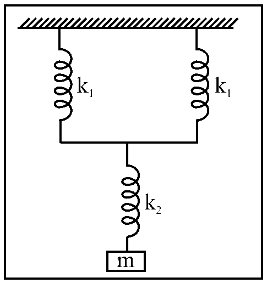
(1) 15 J

(2) 30 J

(3) 7.5 J

(4) 60 J

23. The total spring constant of the system as shown in the figure will be :



(1) $\frac{k_1}{2} + k_2$ (2) $\left[\frac{1}{2k_1} + \frac{1}{k_2}\right]^{-1}$

(3) $\frac{1}{2k_1} + \frac{1}{k_2}$ (4) $\left[\frac{2}{k_1} + \frac{1}{k_2}\right]^{-1}$

24. A particle is projected in a plane perpendicular to a uniform magnetic field. The area bounded by the path described by the particle is proportional to :

(1) Velocity (2) Momentum

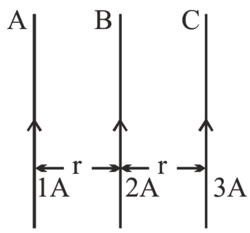
(3) Kinetic energy (4) None of above

25. An electron is moving in a zero gravity region towards vertically upward direction & magnetic field in this region is along east direction. Then what will be the direction of electric field in this region so that electron can remain undeflected :-

(1) North (2) South

(3) East (4) West

26. In the figure A, B and C, are current carrying wires. The direction of resultant force on the wire B will be :-



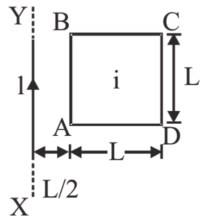
(1) In the plane of paper and towards the right

(2) Perpendicular to plane, out of paper

(3) In the plane of paper and towards the left

(4) Along the direction of current

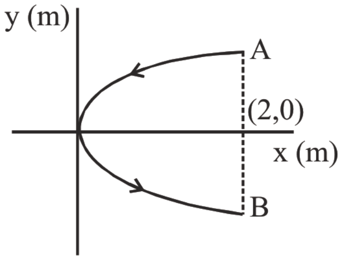
27. A square loop ABCD, carrying a current i, is placed near and coplanar with a long straight conductor XY carrying a current I, the net force on the loop will be :-



(1) $\frac{2\mu_0 Ii}{3\pi}$ (2) $\frac{\mu_0 Ii}{2\pi}$

(3) $\frac{2\mu_0 IiL}{3\pi}$ (4) $\frac{\mu_0 IiL}{2\pi}$

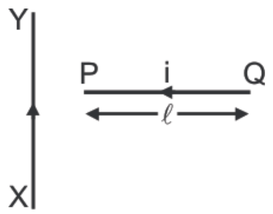
28. A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current I = 2A as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4\hat{k}$ Tesla. The magnetic force on the wire is (in newton)



(1) 16N (2) 64N

(3) 32N (4) 20N

29. A wire PQ carries a current 'i' is placed perpendicular to a long wire XY carrying a current I. Then force between these two wires :-



(1) Attraction

(2) Repulsion

(3) Both (1) and (2)

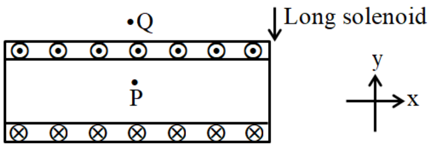
(4) Does not obey newton's third law of motion.

30. For a current carrying long cylinder, magnetic field at the point at distance R/4 is 2 mT then magnetic field at distance 4R is - (distances are from axis and R is radius of cylinder)

(1) 2 mT (2) 1 mT

(3) 4 mT (4) 8 mT

31. A section of solenoid is shown in figure then direction of magnetic field is -



(1) At P in $(+\hat{i})$ direction, at Q in $-\hat{i}$ direction

(2) At P in $(+\hat{i})$ direction, at Q, $B_Q = 0$

(3) At P in $(-\hat{i})$ direction, at Q in + i direction

(4) At P in $(-\hat{i})$ direction , at Q, $B_Q = 0$

32. TOKAMAK uses :

(1) Toroidal field

(2) solenoidal field

(3) Field of long current carrying cylinder

(4) Helmholtz coil's field

33. When charge particle moves in magnetic field applied perpendicular to it's velocity then, magnetic field does not change it's

(1) velocity

(2) momentum

(3) acceleration

(4) angular momentum

34. In phenomenon of 'Aurora' green colour is due to -

(1) Nitrogen

(2) Oxygen

(3) helium

(4) hydrogen

35. An electron moving along circular path with the frequency 10^{10} rps then applied magnetic field is

(1) 0.9 T (2) 0.18 T

(3) 0.27 T (4) 0.36 T

SECTION-B (PHYSICS)

36. An electron of mass m when accelerated through a potential difference V has de-broglie wavelength λ. The de-broglie wavelength associated with a proton of mass M accelerated through the same potential difference will be -

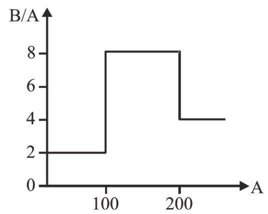
(1) $\lambda\frac{m}{M}$ (2) $\lambda\sqrt{\frac{m}{M}}$ (3) $\lambda\frac{M}{m}$ (4) $\lambda\sqrt{\frac{M}{m}}$

37. Binding energy per nucleon of ${}_3Li^7$ is approximately equal to 5.6 MeV. Equivalent mass defect is :-

(1) 0.018 amu (2) 0.042 amu

(3) 0.006 amu (4) 0.024 amu

38. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below :



(A) Fusion of two nuclei with mass numbers lying in the range of 1 < A < 50 will release energy

(B) Fusion of two nuclei with mass numbers lying in the range of 51 < A < 100 will release energy

(C) Fission of a nucleus lying in the mass range of 100 < A < 200 will release energy when broken into two equal fragments

(D) Fission of a nucleus lying in the mass range of 200 < A < 260 will release energy when broken into two equal fragments :

(1) A & B (2) A & D

(3) B & D (4) C & D

39. An atomic power nuclear reactor can deliver 300 MW. The energy released due to fission of each nucleus of uranium atom U^{238} is 170 MeV. The number of uranium atoms fissioned per hour will be:-

(1) 30×10^{25} (2) 4×10^{22}

(3) 10×10^{20} (4) 5×10^{15}

40. A radioactive nucleus of mass M emits a photon of frequency ν and the nucleus recoils. The recoil energy will be :-

(1) $Mc^2 - h\nu$ (2) $h^2\nu^2 / 2Mc^2$

(3) Zero (4) $h\nu$

41. A sound has an intensity of 2×10^{-8} W/m^2 . Its intensity level (in decibel) is : ($\log_{10} 2 = 0.3$):-

(1) 23 (2) 3 (3) 43 (4) 4.3

42. A particle of mass m performs SHM along a straight line with frequency f and amplitude A :-

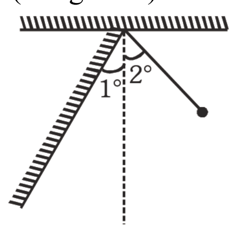
(1) The average kinetic energy of the particle is zero

(2) The average potential energy is $m\pi^2f^2A^2$

(3) The frequency of oscillation of kinetic energy is f

(4) Velocity function leads acceleration by $\pi/2$

43. A simple pendulum of length 1m is allowed to oscillate with amplitude 2°. It collides elastically with a wall inclined at 1° to the vertical. Its time period will be : (use $g = \pi^2$)



(1) 2/3 sec

(2) 4/3 sec

(3) 2 sec

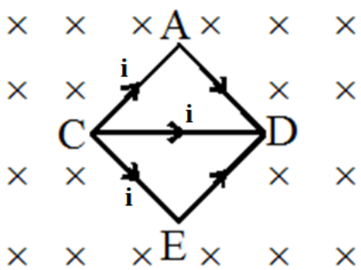
(4) None of these

44. The amplitude of a damped oscillator becomes one third in 2 sec. If its amplitude after 6 sec is 1/n times the original amplitude then the value of n is

(1) 3^2 (2) $3\sqrt{2}$

(3) $3\sqrt{3}$ (4) 3^3

45. Same current i = 2A is flowing in a wire frame as shown in figure. The frame is a combination of two equilateral triangles ACD and CDE of side 1m. It is placed in uniform magnetic field B = 4T acting perpendicular to the plane of frame. The magnitude of magnetic force acting on the frame is



(1) 24 N

(2) Zero

(3) 16 N

(4) 8 N

46. When dielectric is taken out of the charged capacitor then

(1) Capacitance increases

(2) Potential increases

(3) Electric field decreases

(4) Charge increases

47. Which of the following ray is not deflected by the magnetic field

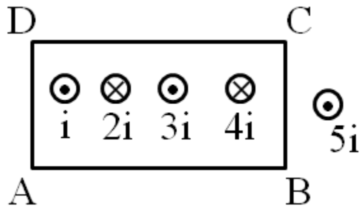
(1) α-rays

(2) β-rays

(3) X-rays

(4) Cosmic rays

48. Calculate $\oint\limits_{ABCDA} B_t d\ell$ for the given loop ABCDA, where B_t represents component of the magnetic field tangential to the loop element :



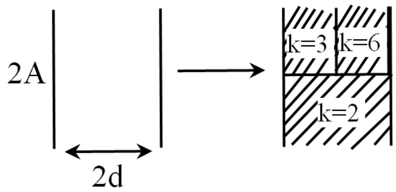
(1) $-\mu_0 i$

(2) $2\mu_0 i$

(3) $-2\mu_0 i$

(4) $\mu_0 i$

49. Equivalent dielectric constant of the given capacitor filled with three dielectrics



(1) 6

(2) 4

(3) 3

(4) 2

50. A charge particle (+q), moving with the velocity $\vec{v} = 2\hat{i} + 3\hat{j}$ m/s enters in a magnetic field $\vec{B} = 0.1\hat{i}$ T from origin, then

(1) It's path remains straight line

(2) It's path will be a circle

(3) 'Z' coordinate of its motion will never be positive

(4) 'Y' co-ordinate of its motion will never be negative

Topic : GOC-I (FULL), TAUTOMERISM, MOLE CONCEPT, CHEMICAL EQUILIBRIUM, [BACK UNIT] : SOLUTION

SECTION-A (CHEMISTRY)

51. 10 moles of CH_4 reacts with 21 moles of O_2 at STP. The number of moles of reactant remains unreacted after the reaction:-

(1) 1 mol O_2 (2) 2 mol O_2

(3) 1 mol CH_4 (4) 2 mol CH_4

52. 10 g of metal chloride contains 8.35 g chlorine and vapour density of its chloride is 85, what is the formula of metal chloride ?

(1) MCl_4 (2) MCl_3

(3) MCl_2 (4) MCl_1

53. A 1000 gram sample of NaOH contains 3 moles of O atoms. What is the % purity of NaOH ?

(1) 14% (2) 100%

(3) 12% (4) 24%

54. The mass of CO_2 that shall be obtained by heating 10 kg of 80% pure lime stone ($CaCO_3$) is :-

(1) 4.4 kg (2) 6.6 kg

(3) 3.52 kg (4) 8.8 kg

55. Number of electrons present in 6 mg of CO_3^{-2} are :-

(1) 1.92×10^{21}

(2) 1.92×10^{20}

(3) 1.92×10^{22}

(4) 2×10^{-3}

56. If 4 mol of NO is taken initially in a 2L vessel and degree of dissociation is 20%, the value of K_P for the reaction $2NO \rightleftharpoons N_2 + O_2$:-

(1) $\frac{1}{(18)^2}$ (2) $\frac{1}{(8)^2}$

(3) $\frac{1}{16}$ (4) $\frac{1}{32}$

57. Arrange the following solutions in order of increasing osmotic pressure. Assume 100% ionisation for electrolytes :-

(I) 1M NaCl

(II) 1M Na_2SO_4

(III) 1M Na_3PO_4

(1) I > II > III (2) III > II > I

(3) II > III > I (4) II > I > III

58. What is molarity of 30% (w/W) urea solution if density of solution is 1.2 g/ml :- (Urea is NH_2CONH_2)

(1) 6 (2) 2 (3) 3 (4) 4

59. Which of the following 0.10 m aqueous solution will have the lowest freezing point ?

(1) $Al_2(SO_4)_3$ (2) $C_5H_{10}O_5$

(3) KI (4) $C_{12}H_{22}O_{11}$

60. The volume strength of a sample of H_2O_2 is 10.08 V. The mass of H_2O_2 present in 250 ml of this solution is :-

(1) 15.3 g

(2) 7.65 g

(3) 3.6 g

(4) 30.6 g

61. ***Assertion :–*** On opening a sealed soda bottle dissolved carbon dioxide gas escapes out.

Reason :– Gas escapes to reach the new equilibrium condition of lower pressure.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

62. If two substances A and B have $P_A^\circ : P_B^\circ = 1 : 2$ and have moles ratio in solution 1 : 2 then mole fraction of A in vapour phase will be :-

(1) 0.33 (2) 0.25

(3) 0.20 (4) 0.52

63. The following concentrations were obtained for the formation of NH_3 from N_2 and H_2 at equilibrium at 500K. $[N_2] = 2 \times 10^{-2}$ M, $[H_2] = 4 \times 10^{-2}$ M and $[NH_3] = 2 \times 10^{-2}$ M. Find equilibrium constant.

(1) 3.12 (2) 312 (3) 156 (4) 624

64. For the equilibrium $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$ the value of equilibrium constant, K_c is 3×10^{-4} at 1000 K. Find K_p for the reaction at this temperature?

(1) 2.5 (2) 0.4 (3) 0.025 (4) 0.04

65. For the reversible reaction,
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ + heat
The equilibrium shifts in the forward direction:

(1) by increasing the concentration of $NH_3(g)$

(2) by decreasing the pressure

(3) by decreasing the concentration of $N_2(g)$ and $H_2(g)$

(4) by increasing pressure and decreasing temperature.

66. The freezing point depression of 0.001 m $K_x[Fe(CN)_6]$ is 7.44×10^{-3} K. Determine value of x (Assume 100% ionisation of salt) [k_f = 1.86 K. Kg/mol for water]

(1) 4 (2) 3 (3) 2 (4) 1

67. At a certain temperature, the equilibrium constant (K_c) is 16 for the reaction :
$SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$
If we take one mole of each of the four gases in one litre vessel, what would be the equilibrium concentration of NO and NO_2 respectively?

(1) 1.6, 1.4 (2) 0.4, 1.6

(3) 1.6, 0.4 (4) 0.4, 0.4

68. Match the column:-

	Column-I		Column-II
A	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ (ΔH = –ve)	P	K increases with increase in temperature.
B	$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ (ΔH = +ve)	Q	K decreases with increase in temperature.
C	$A(g) + B(g) \rightleftharpoons 2C + D(g)$ (ΔH = +ve)	R	Pressure has no effect.
D	$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (ΔH = +ve)	S	Moles of product increase due to addition of inert gas at constant pressure.

(1) (A)⟶Q; (B)⟶P,R; (C)-Q,S; (D)-P,Q

(2) (A)⟶Q; (B)⟶P,R; (C)-P,S; (D)-P,S

(3) (A)⟶P; (B)⟶Q; (C)-R; (D)-S

(4) (A)⟶Q; (B)⟶P,S; (C)-P,S; (D)-R

69. **Statement-I** : For the physical equilibrium $H_2O(s) \rightleftharpoons H_2O(\ell)$ on increasing temperature and increasing pressure more water will form.
Statement-II : Since forward reaction is endothermic in nature and volume of water is greater than that of volume of ice.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

70. Which one of the following is not correct for an ideal solution?

(1) It must obey Raoult's Law

(2) $\Delta H = 0$

(3) $\Delta V = 0$

(4) $\Delta H = \Delta V \neq 0$

71. Mole fraction of a solute in benzene is 0.2. The molality of solute is :-

(1) 3.2 (2) 2 (3) 4 (4) 3.57

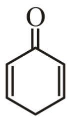
72. The solution which show large positive deviation from Raoult's law forms :-

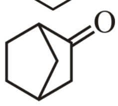
(1) Maximum boiling azeotrope at a specific composition.

(2) Maximum freezing azeotrope at a specific composition.

(3) Minimum boiling azeotrope at a specific composition.

(4) Minimum freezing azeotrope at a specific composition.

73. (A) $CH_3-CH_2-\overset{O}{\overset{||}{C}}-CH_3$ (B) 

(C) $Ph-\overset{O}{\overset{||}{C}}-H$ (D) 

(E) Ph–CH=CH–OH

Which of above can show tautomerism.

(1) A, B, D only (2) A, B only

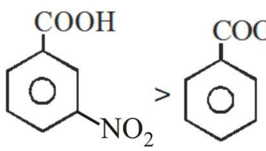
(3) A, B, D, E only (4) All

74. Which of the following has lowest pK_a value.

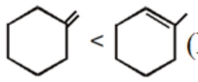
(1) Methoxy acetic acid (2) Acetic acid

(3) Chloro acetic acid (4) Trifluoroacetic acid

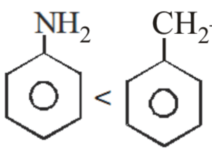
75. Incorrect match is :-

(1) 

(Acidic Strength)

(2) 

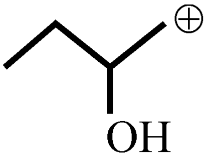
(Heat of hydrogenation)

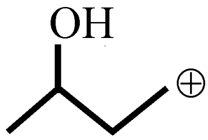
(3) 

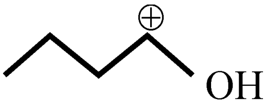
(Basic Strength)

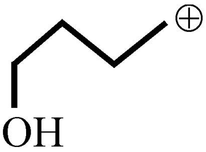
(4) $CH_3-\overset{\ominus}{C}H_2 > CH_3-\overset{\ominus}{\underset{CH_3}{\underset{|}{C}}}-CH_3$ (Stability of anion)

76. Compare stability order of given carbocation.

i. 

ii. 

iii. 

iv. 

(1) i > ii > iii > iv (2) iv > ii > i > iii

(3) iii > iv > ii > i (4) iii > iv > i > ii

77. The strongest base is :

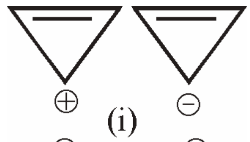
(1) $CH_3-\underset{CH_3}{\underset{|}{N}}-CH_3$

(2) $H_2N-\underset{NH_2}{\underset{|}{C}}=NH$

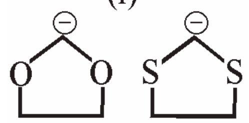
(3) $C_6H_5-\underset{C_6H_5}{\underset{|}{N}}-C_6H_5$

(4) $CH_3-NH-CH_3$

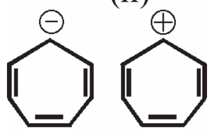
78. In which pair second is more stable than first ?



(i)



(ii)



(iii)

$CH_3O-\overset{\oplus}{C}H_2$ $CH_3-O-CH_2-\overset{\oplus}{C}H_2$

(iv)

(1) (i) and (ii)

(2) (ii) and (iii)

(3) (i), (ii) and (iii)

(4) (i), (ii), (iii) and (iv)

79. **Assertion :-** K_{a_1} of fumaric acid is greater than maleic acid.

Reason :- Conjugate base of maleic acid is more stable due to intramolecular H-bonding.

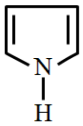
(1) Both Assertion & Reason are correct & the Reason is the correct explanation of the Assertion.

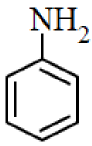
(2) Both Assertion & Reason are correct but Reason is not the correct explanation of the Assertion.

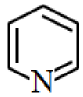
(3) Assertion is correct but Reason is incorrect.

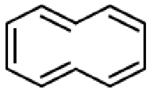
(4) Assertion is incorrect but Reason is correct.

80. Which is aromatic and homocyclic ?

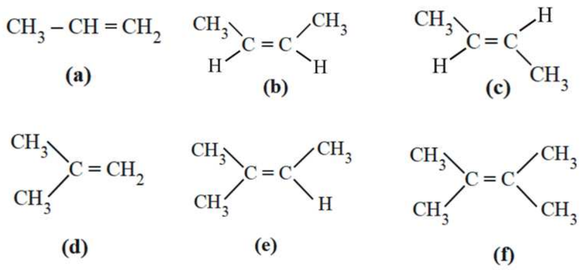
(1) 

(2) 

(3) 

(4) 

81. Consider



Correct order of stability.

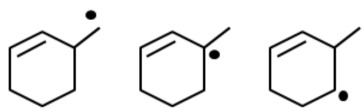
(1) a > b > c > d > e > f

(2) f > e > c > b > d > a

(3) f > e > c > d > b > a

(4) f > e > d > c > b > a

82. Which is correct stability order for following free radical.



(A) (B) (C)

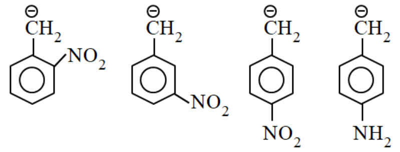
(1) A > B > C

(2) B > A > C

(3) B > C > A

(4) A > C > B

83. Consider



(A) (B) (C) (D)

Correct order of stability.

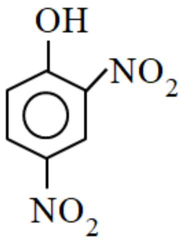
(1) A > B > C > D

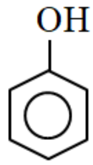
(2) A > C > B > D

(3) C > B > A > D

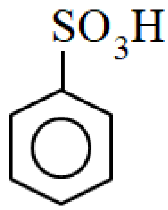
(4) C > A > B > D

84. Which is not soluble in $NaHCO_3$ solution-

(1) 

(2) 

(3) CH_3COOH

(4) 

85. **Assertion (A) :** Acidic strength order

$$R-CH_2-OH > \begin{matrix}R\\ \\R\end{matrix}\!\!>\!CH-OH > R\!-\!\!\!\begin{matrix}R\\ \\R\end{matrix}\!\!>\!C-OH$$

Reason (R) : More number of Alkyl group increase electron density on oxygen atom hence O–H bond will be less polar so acidic strength decreases.

(1) Both Assertion & Reason are correct & the Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

SECTION-B (CHEMISTRY)

86. For the reaction $H_{2(g)} + I_{2(g)} \longrightarrow 2HI_{(g)}$ if 2 L of H_2 react with 4 L of I_2 then volume of HI formed at NTP?

(1) 4 L (2) 2 L (3) 8 L (4) 22.4 L

87. If water sample are taken from sea, river or lake, they will be found to contain hydrogen and oxygen in the ratio of 1:8 by mass. This indicates the law of :-

(1) Multiple proportion (2) Definite proportion

(3) Avogadro law (4) None of these

88. Determine the empirical formula of kelvar used in making bullet proof vest whose percentage composition by mass is 70.6 % C, 4.2 % H, 11.8 % N and 13.4 % O.

(1) $C_7H_5NO_2$ (2) $C_7H_5N_2O$

(3) C_7H_9NO (4) C_7H_5NO

89. What will be the molality of a solution of glucose in water which is 10% w/W ?

(1) 0.01 m (2) 0.617 m

(3) 0.668 m (4) 1.623 m

90. Hydrolysis of sucrose gives,

Sucrose + $H_2O \rightleftharpoons$ Glucose + Fructose

Equilibrium constant K_c for the reaction is 2×10^{10} at 500 K find ΔG^0 at 500 K?

(1) –98.6 kJ/mol (2) –98.6 J/mol

(3) –0.97 kJ/mol (4) –23.7 kJ/mol

91. Consider the following gaseous equilibrium with equilibrium constant K_1 and K_2 respectively

$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$

$2SO_3(g) \rightleftharpoons 2SO_2 + O_2(g)$

The equilibrium constant are related as:

(1) $K_1^2 = \frac{1}{K_2}$ (2) $2K_1 = K_2^2$

(3) $K_2 = \frac{2}{K_1^2}$ (4) $K_2^2 = \frac{1}{K_1}$

92. Match the column :-

	Column-I		Column-II
(A)	0.1 M $Al_2(SO_4)_3$	(P)	Solution with highest Boiling point.
(B)	0.1 M $AlPO_4$	(Q)	Van't Hoff factor is greater than 1
(C)	0.1 M Urea	(R)	Solution with lowest osmotic pressure
(D)	0.1 M $MgCl_2$	(S)	Solution with lowest freezing point

(1) (A)→P,S; (B)→Q; (C)→S; (D)→R

(2) (A)→P,Q,S; (B)→R; (C)→Q; (D)→S

(3) (A)→P,Q,S; (B)→Q; (C)→R; (D)→Q

(4) (A)→P; (B)→Q; (C)→R; (D)→S

93. **Assertion (A) :** People taking a lot of salty food, experience the puffiness or swelling called edema.

Reason (R) : There is water retention in tissue cell and intercellular spaces because of osmosis.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

94. Which of the following has least freezing point?

(1) 1% sucrose (2) 1% NaCl

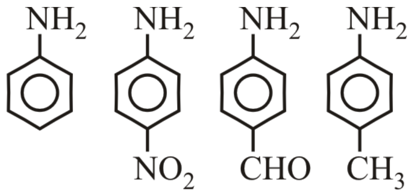
(3) 1% $CaCl_2$ (4) 1% glucose

95. The degree of dissociation of $Ca(NO_3)_2$ in a dilute aqueous solution, Containing 7g of the salt per 100 g of water at 100°C, is 70%. If the vapour pressure of water at 100°C is 760 mm, then vapour pressure of the solution is (nearly) :-

(1) 760 mm (2) 754 mm

(3) 700 mm (4) 746 mm

96. Arrange in decreasing order of basic strength:



(a) (b) (c) (d)

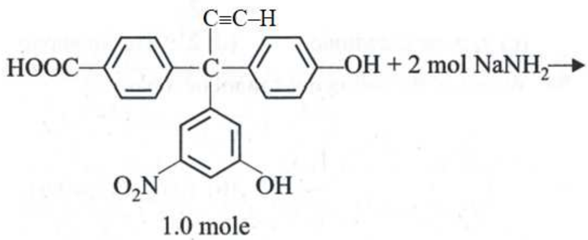
(1) a > b > c > d

(2) b > a > c > d

(3) d > a > c > b

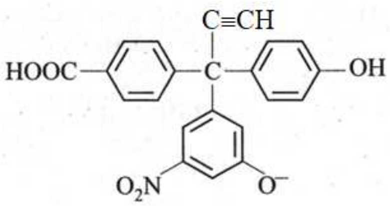
(4) d > c > b > a

97.

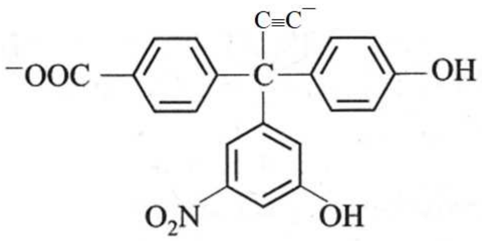


What is formed in the above reaction?

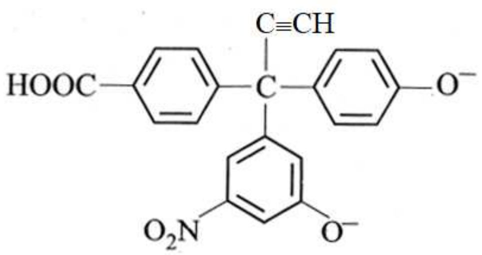
(1)



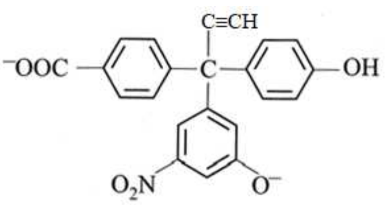
(2)



(3)



(4)



98. **Assertion** :- Alkylamine is more basic then arylamine.

Reason :- Alkylamine have localised lone pair on nitrogen.

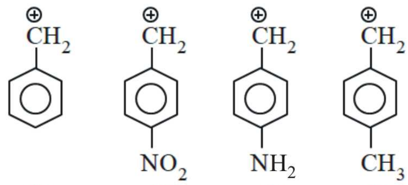
(1) Both Assertion & Reason are correct & the Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Assertion is incorrect but Reason is correct.

99. Consider



(A) **(B)** **(C)** **(D)**

Correct order of stability.

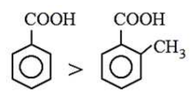
(1) A > B > C > D

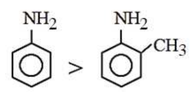
(2) B > A > D > C

(3) C > A > D > B

(4) C > D > A > B

100. Consider following order

(A)  (Acidic strength order)

(B)  (Basic strength order)

(C) $CH_2(Cl)-COOH > HCOOH > CH_2(Cl)-CH_2-COOH$ (Acidic strength order)

(D) $Ph-CH_2-NH_2 > NH_3 > Ph-NH_2$ (Basic strength order)

Which of above are correct order

(1) A, B, C, D

(2) A, B, D

(3) B, C, D

(4) B, D

Topic : BIOTECHNOLOGY, MICROBES IN HUMAN WELFARE, PLANT BREEDING, BIOMOLECULES, PLANT PHYSIOLOGY UPTO MINERAL NUTRITION (TILL DATE), PLANT ANATOMY, ANIMAL HUSBANDRY, [BACK UNIT] : ORIGIN EVOLUTION

SECTION-A (BIOLOGY-I)

101. Match the column - I with column - II and select correct option from table ?

	Column - I		Column - II
(a)	Heart wood	(i)	Conducting element
(b)	Sap wood	(ii)	Mechanical support
(c)	Cork	(iii)	Phellogen
(d)	Cork cambium	(iv)	Phellem

(1) a – i, b – ii, c – iii, d – iv

(2) a – i, b – ii, c – iv, d – iii

(3) a – ii, b – i, c – iv, d – iii

(4) a – ii, b – i, c – iii, d – iv

102. Arrange the following different structures from outside to inside in a tree which are produced during secondary growth?

(a) Wood (b) Secondary phloem

(c) Phellem (d) Phelloderm

(1) c ⟶ d ⟶ a ⟶ b (2) a ⟶ b ⟶ d ⟶ c

(3) c ⟶ d ⟶ b ⟶ a (4) d ⟶ c ⟶ b ⟶ a

103. **Assertion :** In stems, the primary xylem is endarch.

Reason : In stems the protoxylem lies towards the centre (Pith) and metaxylem lies towards the periphery.

(1) Both Assertion and Reason are true and the reason is correct explanation of assertion.

(2) Both Assertion and Reason are true but reason is not correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false

104. Lenticels are :-

(1) Scars on old stem

(2) Aerating pores in bark

(3) Special stomata in hydrophytic plants

(4) Special stomata in leaves

105. What is a position of xylem in the vascular bundle of dorsiventral leaf?

(1) Abaxial (2) Adaxial

(3) Central (4) Dorsal

106. Which of the following is made up of dead cells only ?

(1) Xylem parenchyma (2) Collenchyma

(3) Phellem (4) Phloem

107. Match the following

			Cranial capacity		Character
A	*Homo habilis*	i	1400 cc	a	First tool maker
B	Modern man	ii	700 cc	b	Buried their dead
C	Neanderthal man	iii	900 cc	c	Started agriculture
D	*Homo erectus*	iv	1450 cc	d	First to use fire

(1) A–ii–a ; B–i–c ; C–iv–b ; D–iii–d

(2) A–ii–a ; B–iv–c ; C–i–b ; D–iii–d

(3) A–ii–d ; B–iv–c ; C–i–b ; D–iii–a

(4) A–iii–a ; B–i–c ; C–iv–b ; D–ii–a

108. Select the correct sequence of evolution :-

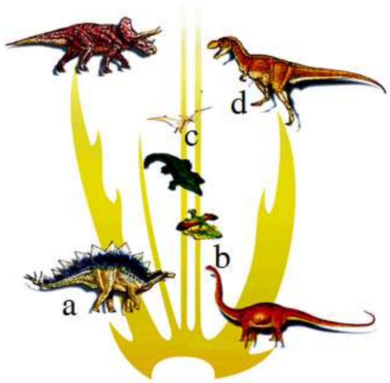
(1) Sauropsid ⟶ thecodont ⟶ lizard

(2) Synapsid ⟶ pelycosaurs ⟶ bird

(3) Sauropsid ⟶ thecodont ⟶ crocodile

(4) Synapsid ⟶ thecodont ⟶ mammals

109. Recognise the figure and find out the correct matching :



(1) a–Archaeopteryx, b–Tyrannosaurs, c–Stegosaurs, d–Pteranodon

(2) d–Archaeopteryx, c–Tyrannosurs, a–Stegosaurs, b–Pteranodon

(3) c–Archaeopteryx, d–Tyrannosaurs, b–Stegosaurs, a–Pteranodon

(4) b–Archaeopteryx, d–Tyrannosaurs, a–Stegosaurs, c–Pteranodon

110. **Statement-I :** Evolution is not a directed process in the sense of determinism.

Statement-II : Evolution is a stochastic process based on chance events in nature and chance mutation in the organisms.

(1) Both Statement-I and Statement-II are incorrect

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I incorrect and Statement-II is correct

(4) Both Statement-I and Statement-II are correct

111. **Assertion :** Lichens can be used as pollution indicators

Reason : Lichens will not grow in areas that are pollution free

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

112. Industrial melanism is an example of:

(1) Mutation (2) Neo Lamarckism

(3) Neo Darwinism (4) Natural selection

113. **Statement-I :** By the time of 500 bya, invertebrates were formed and active.

Statement-II : Plants were widespread on land when animals invaded land.

Choose the correct options from the given statements.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) Both **Statement I** and **Statement II** are correct.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

114. Fossils of which of the following were discovered in 1891?

(1) *Dryopithecus* (2) *Ramapithecus*

(3) *Homohabilis* (4) *Homo erectus*

115. **Statement-I :** The story of evolution of modern man is most interesting and appears to parallel evolution of human brain and language.

Statement-II : Homology is accounted for by the idea of branching descent.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) Both **Statement I** and **Statement II** are correct.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

116. The work of which of the following scientist on populations influenced Darwin?

(1) Hugo devries (2) Thomas Malthus

(3) Lamarck (4) Wallace

117. **Assertion (A) :** Evolution by natural selection, in a true sense would have started when cellular forms of life with differences in metabolic capability originated on earth.

Reason (R) : The rate of appearance of new forms is linked to the life cycle or the life span.

Choose the correct answer from the given options :-

(1) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is correct but **Reason** is not correct.

(3) Both **Assertion** and **Reason** are incorrect.

(4) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.

118. **Assertion :** When we look at stars on a clear night sky we are, in a way, looking back in time.

Reason : The emitted light from these stars started it's journey millions of years back and reaching our eyes now.

Choose the correct answer from the given options.

(1) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is correct but **Reason** is not correct.

(3) Both **Assertion** and **Reason** are incorrect.

(4) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.

119. Which pair is not matched correctly?

(1) Land reptiles ⟶ Dinosaurs

(2) Fish like reptiles ⟶ *Ichthyosaurs*

(3) Biggest dinosaur⟶ *Triceratops*

(4) 1^{st} mammals ⟶ Like shrews

120. Which of the following were existing 15 mya?

(1) *Dryopithecus* (2) *Homo erectus*

(3) *Ramapithecus* (4) Both (1) and (3)

121. In the given list how many are placental mammals?

Numbat, Flying phalanger, Tasmanian wolf, Spotted cuscus, Flying squirrel.

(1) 4 (2) 3 (3) 2 (4) 1

122. Which pair is not matched correctly?

(1) Agriculture came ⟶ 10,000 years ago

(2) Prehistoric cave art ⟶ 18,000 years ago

(3) Modern *Homo sapiens* arose ⟶ 1.5 mya

(4) Homo habilis ⟶ 1st human like being

123. Which of the following statement is false?

(1) Lobefins evolved into first amphibians.

(2) Coupled to enhance reproductive success, natural selection can make it look like a different population.

(3) Microbial experiments show that pre-existing advantageous mutations when selected will result in observation of new phenotypes.

(4) If gene migration happens multiple times it is called founder effect.

124. Which of the following is not derived from Psilophyton?

(1) Rhynia type plant (2) Sphenopsids

(3) Progymnosperms (4) Ferns

125. Origin of universe can be explained by?

(1) Panspermia theory

(2) Stanley miller theory

(3) Big bang theory

(4) Spontaneous generation theory

126. Which pair is not matched correctly?

(1) Wings of butterfly and birds ⟶ Analogy

(2) Flippers of Penguin and Dolphin ⟶ Analogy

(3) Forelimbs of mammals ⟶ Divergent evolution

(4) Eyes of octopus and mammals ⟶ Divergent evolution

127. Which of the following statement is not correct?

(1) The biochemical similarities point to the same shared ancestry as structural similarities among diverse organisms.

(2) Wombat is an example of marsupial radiation.

(3) When we describe the story of this world we always describe evolution as a result of a process.

(4) The essence of Darwinian theory about evolution is natural selection.

128. Which of the following is the viral disease in birds ?

(1) Bird flu

(2) Ringworm

(3) New castle disease (Ranikhet)

(4) 1 & 3 both

129. **Assertion:** Apiculture during flowering period increases pollination efficiency and improves the yield.

Reason: Bees are the pollinators of many crop species such as Sunflower, Brassica, apple and pear.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

130. Some statements are given about the MOET. How many statements are correct among these?

(1) In this method, a cow is administered hormones, with FSH-like activity.

(2) Genetic mother produces 6-8 ova per cycle in this method.

(3) The fertilized eggs are transferred into surrogate mothers when they achieve a stage of more than 32 cells.

Options:

(1) Zero (2) One (3) Two (4) Three

131. Fill in the blanks and choose the correct option

(1) Hisardale is a new breed of sheep developed in Punjab by crossing ___(A)____ and ___(B)___.

(2) Continued inbreeding, especially close inbreeding, usually reduces fertility and even productivity. This is called ___(C)____.

(3) ___(D)____ increases homozygosity.

(4) Inbreeding refers to the mating of more closely related individuals within the same breed for ____(E)____ generations.

	(A)	(B)	(C)	(D)	(E)
(1)	Bikaneri rams	Marino ewes	Inbreeding depression	Outbreeding	4-6
(2)	Bikaneri rams	Marino ewes	Inbreeding depression	Inbreeding	4-6
(3)	Bikaneri ewes	Marino rams	Inbreeding depression	Inbreeding	4-6
(4)	Bikaneri ewes	Marino rams	Inbreeding depression	Inbreeding	6-8

132. Select the incorrect statement ?

(1) Fishery has an important role in Indian economy

(2) Bee keeping is a labour-intensive

(3) Milk yield is primarily depends on the quality of breed.

(4) Hisardale is a new hybrid breed of sheep.

133. Select the correct pair of fresh water fish.

(1) Rohu & Sardine (2) Catla & Mrigal

(3) Sardine & Pomfret (4) Catla & Salmon

134. RNA interference (RNAi) technique has been devised to protect the plants from the nematode. In this technique mRNA of nematode is silenced by :-

(1) dsDNA (2) ssDNA

(3) dsRNA (4) Target proteins

135. cry ll Ab and cry l Ab produce toxins that control:-

(1) cotton bollworms and corn borer respectively

(2) corn borer and cotton bollworms respectively

(3) tobacco budworms and nematodes respectively

(4) nematodes and tobacco budworms respectively

SECTION-B (BIOLOGY-I)

136. Pericycle performs how many functions in a dicot root ?

Formation of Lateral roots, formation of Casparian bands, Formation of part of vascular cambium, Formation of Cork cambium, Formation of inter fascicular cambium.

(1) Three

(2) Four

(3) Two

(4) Five

137. Read the following statements.

I. Found in seed coat of legumes.

II. Deposition of pectin occurs.

III Very narrow lumen.

IV Provide mechanical support to petiole.

How many statements from above are true for sclereids (sclerenchyma)?

(1) II, III, IV (2) I, III, IV

(3) I, III (4) I, IV

138. From where does the vascular cambium develops, in dicot root –

(1) Tissue just below the phloem bundles i.e. conjuctive tissue

(2) A portion of pericycle tissue above the protoxylem

(3) Both (1) and (2)

(4) Cells of medullary rays

139. Match the column-I with column-II and choose the correct option :-

	Column-I		Column-II
A	Ist cellular form of life.	i	4500 MYA.
B	Evolution of Jawless fishes.	ii	500 MYA.
C	Disappearance of dinosaurs.	iii	4000 MYA.
D	Formation of Invertebrates.	iv	350 MYA.
E	Appearance of life on earth.	v	2000 MYA.
		vi	65 MYA.
		vii	320 MYA.

(1) A-iii, B-vii, C-vi, D-iv, E-v

(2) A-v, B-iv, C-vi, D-ii, E-iii

(3) A-v, B-ii, C-viii, D-iv, E-i

(4) A-iii, B-ii, C-vi, D-iv, E-v

140. According to most accepted theory of evolution, the life has:

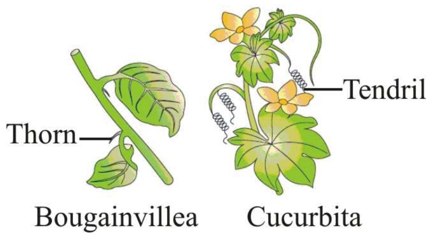
(1) Come from outer space

(2) Come out of dead and decaying matter

(3) Originated from pre-existing non-living organic molecules

(4) Created by god

141. The thorn of *Bougainvillea* and Tendril of *Cucurbita* in the given figure does not show :

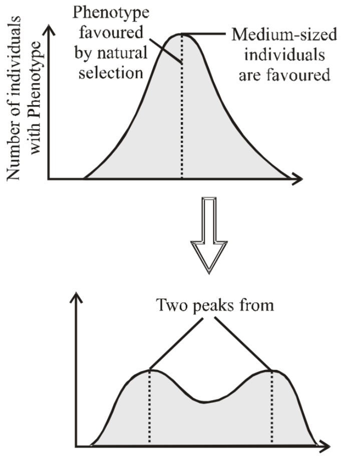


(1) Homology

(2) Divergent evolution

(3) Common ancestry

(4) Vestigial structures

142. 

This diagrammatic representation shows which type of natural selection.

(1) Directional selection

(2) Stabilising selection

(3) Disruptive selection

(4) None of these

143. Choose the incorrect statement:

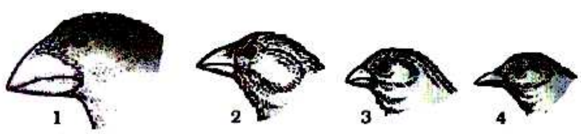
(1) There was no atmosphere on early earth

(2) Life appeared about 500 million years after the formation of earth

(3) The first non-cellular form of life could have been originated about 3 billion years back

(4) The earth is about 4000 years old

144. The diagram given below represents a variety of beaks of finches that Darwin found in Galapagos island?



What type of evolutionary process is depicted in the diagram?

(1) Parallel adaptation

(2) Convergent evolution

(3) Adaptive radiation

(4) Adaptive convergence

145. Which of the following was formed during simulation experiment of Stanley Miller ?

(1) Sugar (2) Fatty acids

(3) Nitrogenous base (4) Amino acids

146. Examples of evolution by Anthropogenic action refers to :

(1) Evolution which occurs in nature

(2) Evolution which took place in short time due to human activities

(3) Only human driven but take more time in comparison to nature always.

(4) Not dependent on human but is due to interspecific struggle

147. Which of the following is most potent force for organic evolution :-

(1) Intra-specific struggle

(2) Inter specific struggle

(3) Environmental struggle

(4) All of above

148. In which one of the following options Column I is correctly matched with Column II?

	Column I	Column II
(1)	Fresh water fish	Sardine
(2)	Marine water fish	Rohu
(3)	Fisheries	Shell fish
(4)	Green revolution	Increase fish production

149. Which of the following disease is caused by virus :

(1) Botulism (2) Bird flu

(3) Babesiosis (4) Ascariasis

150. Mules are produced by -

(1) Inbreeding

(2) Out-crossing

(3) Cross-breeding

(4) Interspecific hybridisation

Topic : BIOTECHNOLOGY, MICROBES IN HUMAN WELFARE, PLANT BREEDING, BIOMOLECULES, PLANT PHYSIOLOGY UPTO MINERAL NUTRITION (TILL DATE), PLANT ANATOMY, ANIMAL HUSBANDRY, [BACK UNIT] : ORIGIN EVOLUTION

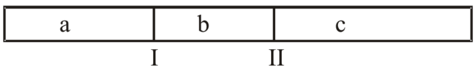
SECTION-A (BIOLOGY-II)

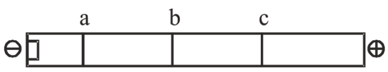
151. If a p^{BR322} vector is cut by ECORI enzyme, then which of its marker gene shows insertional inactivation ?

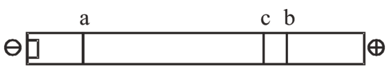
(1) Tet^R gene (2) Amp^R gene

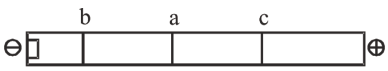
(3) Gal gene (4) None of these

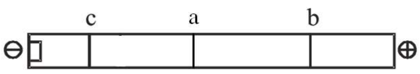
152. The given segment of DNA has restriction sites I and II, which create restriction fragments a, b and c. Which of the following gel produced by electrophoresis would represent the separation & identity of these fragments?



(1) 

(2) 

(3) 

(4) 

153. In pBR 322, a foreign DNA is ligated at Pst-I site, then recombinant plasmids will lose the resistance to the -

(1) Tetracycline (2) Ampicillin

(3) Neomycin (4) Erythromycin

154. Which of the following can't be achieved using PCR?

(1) Detect HIV in AIDS suspect

(2) Detect mutations in cancer patients.

(3) Detect antigen–antibody interaction.

(4) Detect specific micro organisms in sample.

155. The majority of Baculo viruses are used as biological control agents because

(1) These are excellent candidates for species specific applications

(2) They have been showing no negative impact on plants, mammals and non-target insects

(3) Beneficial insects are conserved

(4) All of the above

156. Pusa swarnim variety of *Brassica* is resistant to_______ disease :-

(1) Bacteria blight (2) Black rot

(3) White rust (4) Hill bunt

157. Which one of the following is an example of mutational breeding :-

(1) Parbhani kranti, (YMV resistant variety)

(2) Mungbean (YMV resistant variety)

(3) Pomato

(4) *Saccharum officinarum*

158. Which molecules are found in acid insoluble fraction of biomolecules ?

(1) Polysaccharides and proteins only

(2) Polysaccharides and proteins and nucleic acids only

(3) Proteins and nucleic acids only

(4) Proteins, polysaccharides, nucleic acids and lipids

159. Which of the following is not a similarity between starch and cellulose ?

(1) They are homopolysaccharides

(2) They are found mainly in plants

(3) They form helical structure and hold I_2

(4) They have glycosidic bonds

160. Match the following :-

	Column-I		**Column-II**
a	Alkaloid	I	Vinblastin, curcumin
b	Essential oils	II	Morphine, Codeine
c	Toxins	III	Lemon grass oil
d	Drugs	IV	Abrin, Ricin

(1) a-II, b-III, c-IV, d-I (2) a-III, b-II, c-IV, d-I

(3) a-II, b-III, c-I, d-IV (4) a-III, b-II, c-I, d-IV

161. Match the following

Column-I		Column-II	
(A)	Brown rust of wheat	(i)	Virus
(B)	Tobacco mosaic	(ii)	Bacteria
(C)	Black rot of crucifers	(iii)	Fungi
(D)	Red rot of sugarcane	(iv)	Bollworm

(1) A-(ii), B-(i), C-(iii), D-(iv)

(2) A-(iv), B-(iii), C-(ii), D-(i)

(3) A-(iii), B-(ii), C-(i), D-(iv)

(4) A-(iii), B-(i), C-(ii), D-(iii)

162. Consider the following statements.

I. Solid stem in wheat lead to preference by stem sawfly.

II. In cotton, smooth leaf and presence of nectar do not attract bollworms.

III. In maize, high aspartic acid, low nitrogen and sugar content protect them from stem borers.

Which of the statements given above are incorrect ?

(1) I, II and III (2) I and II

(3) I and III (4) II and III

163. Identify the blank spaces A, B, C and D given in the following table and select the correct answer.

Type of Microbes	Scientific Name	Commercial, Product
Bacterium	A	Lactic acid
Fungus	B	Cyclosporin-A
C	Monascus purpureus	Statins
Fungus	Penicillium notatum	D

(1) A-Lactobacillus, B-Trichoderma polysporum, C-yeast, D-Penicillin

(2) A-Staphylococcus, B-Clostridium C-Yeast, D-Penicillin

(3) A-Lactobacillus, B-Microsporum, C-Yeast, D-Penicillin

(4) A-Staphyloccus, B-Microsporum, C-Agaricus, D-Penicillin

164. Match the items of column-I with column-II and choose correct answer :-

Column-I		Column-II	
(A)	Lady bird	(I)	*Methanobacterium*
(B)	Mycorrhiza	(II)	*Trichoderma*
(C)	Biological control	(III)	Aphids
(D)	Biogas	(IV)	*Glomus*

(1) A – II, B – IV, C – III, D – I

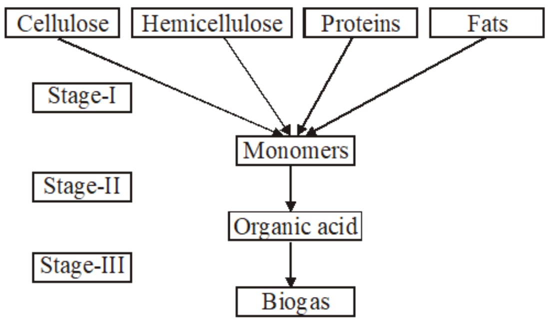
(2) A – III, B – IV, C – II, D – I

(3) A – IV, B – I, C – II, D – III

(4) A – III, B – II, C – I, D – IV

165. Biogas production is a three step anaerobic digestion of animal and other organic wastes.

Study the following flow chart and select the correct option for stage I, II and III :-



(1) In stage-I, facultative anaerobic microbes bring about enzymatic break down of complex organic compounds into simple soluble compound or monomers

(2) In stage-II, monomers are converted into organic acid by fermentation causing microbes

(3) In stage-III, organic acids are acted upon by methanogenic bacteria to produce biogas

(4) All of the above

166. I.... A... is the ability of a cell to take up foreign DNA.

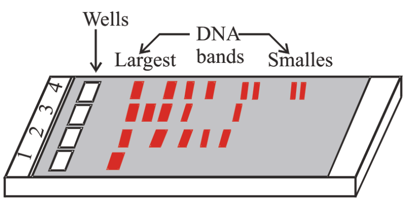
II. The cell is treated with specific concentration of a divalent cation such as ...B... to increase pore size in cell wall.

III. In... C... method recombinant DNA is directly injected into the nucleus of an animal cell.

The most appropriate option regarding A, B and C is

1	A-Competency,	B-Calcium,	C-gene gun method
2	A-Transformation,	B-Sodium,	C-microinjection method
3	A-Competency,	B-Calcium,	C-microinjection method
4	A-Transformation,	B-Sodium,	C-gene gun method

167. Given below is typical agarose gel block in which electrophoresis has been done.



Select the correct statement :-

(1) Lane-1 shows digested DNA fragments

(2) Lane-2 to 4 shows undigested set of DNA fragments.

(3) Sample was loaded in the wells which is placed towards positive electrode

(4) Sample was loaded in the wells which is placed towards negative electrode.

168. Arrange the steps of rDNA technology in correct order:-

I. Extraction of the desired gene product.

II. Amplification of gene of interest.

III. Isolation of desired DNA fragment.

IV. Ligation of DNA fragment into vector.

V. Insertion of rDNA into host.

(1) I, II, III, IV, V (2) V, IV, III, II, I

(3) III, II, IV, V, I (4) III, IV, II, I, V

169. ***Assertion*** :- In gene therapy of SCID, a patient requires periodic infusion of genetically engineered lymphocytes.

Reason :- If the ADA gene is introduced into cells at early embryonic stage, it could be a permanent cure.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

170. Read the following four statements (A-D) :-

(A) The first transgenic buffalo, Rosie produced milk which was human alpha-lactalbumin enriched.

(B) Restriction enzymes are used in isolation of DNA from other macro molecules.

(C) Downstream processing is one of the steps of r-DNA technology.

(D) Disarmed pathogen vectors are also used in transfer of r-DNA into the host.

Which are the two correct statements ?

(1) Statements (A) and (B)

(2) Statements (B) and (C)

(3) Statements (C) and (D)

(4) Statements (A) and (C)

171. Read the following statements and select the correct ones :-

(i) Same kind of sticky ends are produced when vector DNA and DNA with desired gene are cut by different restriction enzymes.

(ii) Exonucleases make cuts at specific positions within the DNA.

(iii) Each restriction endonuclease recognizes a specific palindromic nucleotide sequences in the DNA.

(iv) DNA primers are used during polymerase chain reaction.

(v) Presence of more than one recognition sites of same restriction enzyme within the vector facilitates the gene cloning.

(1) i, iii and iv (2) i and iv

(3) iii and iv (4) ii, iii and iv

172. Which of the following statement is not correct, if a DNA is inserted in plasmid within the coding sequence of an enzyme β-galactosidase?

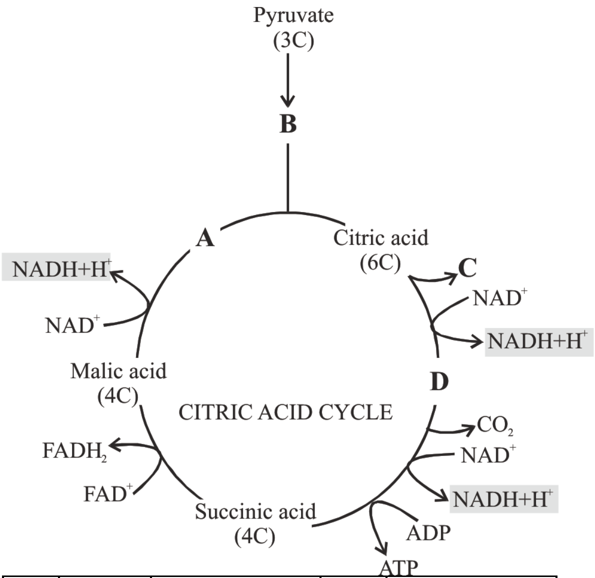
(1) Bacteria having plasmid with insert gives colourless colonies in presence of chromogenic substrate.

(2) It lead to insertional activation of enzyme producing gene.

(3) The ability of enzyme synthesis is lost in the plasmid having insert

(4) Bacteria having plasmid without insert gives blue coloured colonies in presence of chromogenic substrate.

173. In the given representation of Krebs cycle, Identify A, B, C and D



	A	B	C	D
(1)	OAA	Acetyl CoA	CO_2	α-ketoglutaric acid
(2)	OAA	Acetyl CoA	CoA	α-ketoglutaric acid
(3)	Acetyl CoA	OAA	CoA	CO_2
(4)	Acetyl CoA	α-ketoglutaric acid	CO_2	CoA

174. Arrange the following e^- acceptors complex of ETS in inner mitochondrial membrane on the basis of increasing redox potential.

(a) Succinate dehydrogenase

(b) NADH dehydrogenase

(c) Cytochrome a-a_3

(d) Cytochrome b-c_1

(1) (a) (b) (c) (d)
(2) (b) (a) (c) (d)
(3) (b) (d) (a) (c)
(4) (b) (a) (d) (c)

175. Although the aerobic process of respiration takes place only in the presence of oxygen, the role of oxygen is limited to the ____A____ of the process. yet, the presence of oxygen is vital, since it drives the whole process by ____B___ from the system

(1) A = Krebs cycle, B = removing electron

(2) A = Krebs cycle, B = removing hydrogen

(3) A = terminal stage, B = removing hydrogen

(4) A = link reaction and Krebs cycle, B = removing electron

176. Total number of ATP molecules formed during complete oxidation of fructose 1, 6 bisphosphate is :

(1) 20
(2) 32
(3) 36
(4) 40

177. Consider the following about facilitated diffusion and choose the odd one :-

(1) A concentration gradient must already be present for molecules to diffuse

(2) It is sensitive to inhibitors which react with lipid side chain

(3) Water channels are made up of 8 different types of aquaporins

(4) ATPs are not required

178. Consider the following statement

A. Creates transpiration pull for absorption and transport in plants

B. Not supplies water for photosynthesis

C. Transport minerals from the soil to all parts of the plant

D. Cools leaf surfaces, sometimes 10 to 15 degrees, by evaporative cooling

E. Maintain the shape and structure of the plants by keeping cells turgid

Choose the incorrect option w.r.t. transpiration and photosynthesis a compromise:

(1) A and B (2) Only B

(3) A, B, C & D (4) E, A, B

179. Components of water potential of four cells of an actively transpiring wheat plant are :-

Cell	Osmotic potential	Pressure potential
a	–0.9	0.5
b	–0.8	0.6
c	–0.6	0.1
d	–0.7	0.4

Identify the four cells as root hair, cortical cell, endodermal cell and pericycle cell.

(1) a, d, b, c (2) b, c, d, a

(3) b, d, c, a (4) b, d, a, c

180. How many of the following are associated with food translocation in phloem :-

Negative hydrostatic pressure, Sucrose, Active loading, Bulk flow, Passive unloading, Root Pressure

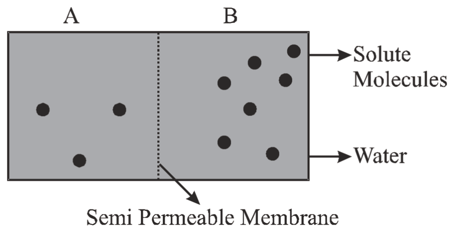
(1) Five

(2) Four

(3) Two

(4) Three

181. Go through the following figure :



Now, point out the incorrect statement :-

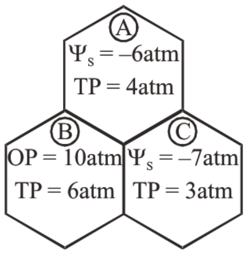
(1) Movement of solvent molecules will take place from chamber A to B

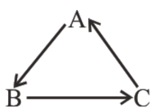
(2) Movement of solute will take place from chamber A to B

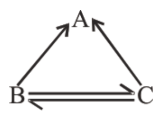
(3) Presence of a SPM is a prerequisite for osmosis to occur

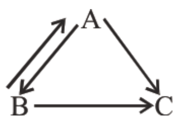
(4) The direction and the rate of osmosis depend upon both the pressure gradient and concentration gradient

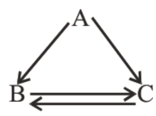
182. Find the direction of movement of water in the adjacent plant cells shown below :-



(1)


(2)


(3)


(4)


183. Mark the incorrect statements

(A) In yeast, incomplete oxidation of glucose is achieved under aerobic conditions, where substrate is converted to CO_2 and ethanol.

(B) Net gain is only 2ATP molecules during fermentation.

(C) Lactic acid is produced in muscles during exercise, when oxygen supply is inadequate for cellular respiration.

(D) The reducing agent during fermentation is $NADPH + H^+$ which is reoxidized to $NADP^+$.

(1) A and D (2) A and B

(3) C and D (4) B and C

184. **Assertion** : For complete breakdown of respiratory substrate krebs cycle is essential.

Reason : Maximum two decarboxylation can occur in each turn of krebs cycle.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

185. Read the following statements (a-d).

(a) Numerically___(A)___is equivalent to the osmotic potential but the sign is opposite.

(b) Conventionally the water potential of ___(B)___ at standard temperature which is not under any pressure is taken to be zero.

(c) Minerals are present in the soil (C)___ which can move across cell membrane.

(d) The movement of water through the root layer is ultimately___(D)___in the endodermis.

Which of the following options is correct w.r.t. blank (A)-(D) in the statements ?

(1) A-Water potential, B-Solution

(2) B-Pure water, D-apoplastic

(3) A-Osmotic pressure, C-Charged particle

(4) Non-charged particles, D-Apoplast

SECTION-B (BIOLOGY-II)

186. Which is not true for cloning vector :-

(1) These are independent of the control of chromosomal DNA

(2) These get integrated into chromosomal DNA after transformation

(3) They contain genes for antibiotic resistance

(4) They have 'ori' region

187. In gel electrophoresis, the separated DNA fragments are visualised after staining the DNA with ...A... followed by exposure to ...B... Here A and B refers to :-

	A	B
1	Ethidium bromide	Infrared radiation
2	Ethidium bromide	UV radiation
3	Ethidium nitrate	γ-rays
4	Ethidium chloride	Radiowave

188. Which of the following is not a direct method of gene transfer in plants?

(1) *Agrobacterium tumefaciens*

(2) Gene gun method

(3) Biolistics

(4) Electroporation

189. Which one is not a biofertilizer ?

(1) *Nostoc* (2) *Mycorrhiza*

(3) *Agrobacterium* (4) *Rhizobium*

190. Which of the following fermented beverage is not produced by distillation ?

(1) Whisky (2) Brandy (3) Rum (4) Wine

191. Virus free plants can be obtained by :-

(1) Only apical meristem culture

(2) Only axillary meristem culture

(3) Apical and axillary meristem culture

(4) Embryo culture

192. The correct order of chemical composition of living tissues/cells in term of % of the total cellular mass :-

(1) H_2O > Nucleic acid > proteins > carbohydrates > lipids > Ions

(2) H_2O > proteins > Nucleic acids > carbohydrates > lipids > Ions.

(3) Nucleic acid > proteins > H_2O > carbohydrates > lipids > Ions

(4) Lipids > ions > carbohydrates > proteins > nucleic acid > H_2O

193. How many plant varieties in the list given below are high yielding wheat varieties developed during green revolution ?
Jaya, Sonalika, Kalyan Sona, Ratna, Sharbati Sonora, Pusa lerma, IR-8, TN-1

(1) Three (2) Four (3) Five (4) Six

194. Biolistics procedure involves :

(1) Making transient pores in the cell membrane to introduce desired gene

(2) Purification of saline water with the help of a membrane system

(3) Opening of stomatal pore during night by artificial light

(4) Plant cells are bombarded with high velocity microparticles of gold / tungsten coated with DNA

195. Fill up the blanks.
(A) When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of _____ and these can be joined together for end-to-end using ______.
(B) Disruption of the cell wall can be achieved by treating the bacterial cells, plant or animal tissue with enzymes such as _____ (bacteria), _____ (plant cells) and _____ (fungus).
(C) If any protein encoding gene is expressed in a heterologous host, it is called as _____.
(D) In gel electrophoresis, the separated DNA fragments can be seen only after staining the DNA with a compound known as _____ followed by exposure to UV radiation.

(1) A-sticky ends, recombinase; B-lysozyme, cellulase, chitinase; C-recombinant protein; D-ethidium bromide

(2) A-sticky ends, DNA ligases; B-exonuclease, cellulase, chitinase; C-recombinant protein; D-ethidium bromide

(3) A-cut ends, DNA topoisomerase; B-hemicellulose, cellulase, chitinase; C-native glycoprotein; D-CSCI

(4) A-sticky ends, DNA ligases; B-lysozyme, cellulase, chitinase; C-recombinant protein; D-ethidium bromide

196. Match the following respiratory substrate with their RQ.

(A) Tripalmitin	(i) Zero
(B) Glucose	(ii) 4
(C) Oxalic acid	(iii) 1
(D) Succulent plant	(iv) 0.7

(1) A(i), B(iii), C(ii), D(ii)

(2) A(i), B(ii), C(iii), D(iv)

(3) A(iv), B(iii), C(i), D(ii)

(4) A(iv), B(iii), C(ii), D(i)

197. Which of the following statement is incorrect?

(1) Entry of water into a cell results in decrease of osmotic pressure of cell.

(2) Higher is the DPD of a cell more is the demand of water

(3) Addition of solute increases osmotic pressure of a solution

(4) When turgor pressure in a plant cell is high osmotic pressure becomes zero

198. A girdled plant may survive for sometime but it will eventually die, because :-

(1) Water will not move downwards

(2) Water will not move upwards

(3) Sugars and other organic materials will not move downwards

(4) Sugars and other organic materials will not move upwards

199. Identify the mis-match in amphibolism of Kreb's cycle :-

(1) Acetyl Co–A $\longrightarrow$ Gibberellic acid

(2) Succinyl Co–A $\longrightarrow$ Cytochromes

(3) α–ketogluteric acid $\longrightarrow$ synthesis of chlorophyll

(4) Oxalo acetic acid $\longrightarrow$ Amino acid

200. If 5 molecule of water are produced in ETS of mitochondria then find the number of ATP produced and $NADH_2$ oxidised :-

(1) 30 ATP, 10$NADH_2$ (2) 10ATP, 5$NADH_2$

(3) 15 ATP, 5$NADH_2$ (4) 6ATP, 6$NADH_2$

SPACE FOR ROUGH WORK

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QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)

ENTHUSE ADV. COURSE : MEA & MEPS

08 10-09-2023	**SYLLABUS COVERED FROM 16-08-2023 TO 09-09-2023**	**P**	MAGNETIC EFFECT OF CURRENT AND MAGNETISM
		C	ELECTROCHEMISTRY
		B	ECOLOGY

TG: @Chalnaayaaar